

Andrew Nash

Staff Autonomous Driving Engineer | Perception | Localization | Planning

@ anash1324@gmail.com 📍 Ottawa, Canada | open to relocation in LinkedIn GitHub Portfolio

Summary. Staff Autonomous Driving Engineer who shipped Canada’s first medium-speed SAE L4 shuttle, Transport-Canada-authorized. Full-stack AV and robotics: machine-learning perception, factor-graph localization, real-time motion planning. Tech lead and code owner of the production stack.

Experience

Aurrigo

Staff ADS Engineer

📅 Oct 2025 - Present 📍 Ottawa, Canada

- **Tech lead of 6 engineers** across navigation, perception, localization, and planning; code owner of two \$1M+ production deployments.
- **Built navigation, perception, localization, and calibration** for Canada’s first medium-speed SAE L4 shuttle; Transport Canada-authorized [press].
- **Shipped PyTorch camera+LiDAR 3D detection + tracking at 83.4 mAP @ 20 Hz**; classical safety channel for ISO 26262 diverse redundancy.
- **Built real-time ROS C++ Frenet motion planner** (replacing legacy PID): 400+ parallel trajectories at 50 Hz, jerk-optimal selection against perception.
- **Cut lateral error 10x (0.5 m to 0.05 m), more than doubled top speed (15 to 35 km/h)**: custom GTSAM factors with dynamic GPS/lidar/IMU sensor fusion; unlocked tunnel + GPS-denied terminals.
- **Authored Aurrigo’s end-to-end Python labelling data pipeline**: annotation tool + auto-labeller lifting 2D foundation models to 3D LiDAR.

ROS C++ PyTorch CUDA TensorRT Git Computer Vision

Aurrigo

ADS Team Lead, North America

📅 Oct 2023 - Oct 2025 📍 Ottawa, Canada

- **Led Aurrigo’s first AV simulator from scratch**: 1000s of scenarios proving airport-customer safety; 50x speedup vs CARLA, faster-than-real-time.
- **Solo-authored Aurrigo’s C++ GTSAM iSAM2 SLAM and mapping** (glim-based): GPU factor graph on gtsam_points, GPS-aligned LiDAR maps, all sensors.
- **Doubled top autonomous speed (25 to 50+ km/h)** via 27x CUDA VGICP scan-match speedup, multi-LiDAR at 20 Hz, and custom Ackermann + crab-steer kinematic factors, on the same embedded hardware.

ROS C++ Python GTSAM UE4 CI/CD Linux Bash

Internships

Nasdaq (Verafin) | Machine Learning Engineer

May - Aug 2021

Lifted gradient-boosting fraud-detection model precision by 6 pp at fixed recall.

Python Pandas Scikit-Learn XGBoost LightGBM AWS

Bank of Canada | Data Science Developer

Sept 2020 - Apr 2021

Built anomaly-detection models automating economist data-review workflows.

Python Pandas Scikit-Learn XGBoost JavaScript Tableau

IBM | NLP Developer

Jan - Apr 2020

Improved enterprise semantic matching by 8% via weighted word embeddings.

Python NLP Keras NLTK Java JavaScript Kubernetes Cognos

Royal Bank of Canada | Full Stack Data Scientist

May - Aug 2019

Built full-stack semantic search prototype with NLP backend and web frontend.

Python NLP BERT Flask React JavaScript Docker

BlackBerry QNX | NLP Developer

Sept - Dec 2018

Applied NLP classifiers to automate software-license tagging across QNX repos.

Python NLP Keras FastText QNX Docker MongoDB

Education

MSc Computer Science (Thesis)

Memorial University Of Newfoundland

📅 2021 - 2023 📍 NL, Canada

4.0 GPA. Thesis: Herds Eye View (AIIDE-2023).

BEng Computer Engineering (Co-Op)

Memorial University Of Newfoundland

📅 2016 - 2021 📍 NL, Canada

Capstone: SR AUV (best 2021 capstone).

Publications

Herds Eye View

Accepted to AIIDE-2023

🔗 arXiv 🔗 Blog Post

Unity RL Transformer PyTorch

Projects

Sage | Real-time D&D AI Co-pilot

- Streaming audio to LLM agent at 1s end-to-end latency: Deepgram STT, Gemini agent loop, 20+ MCP tools.
- Solo-architected: 98 REST routes, hybrid RAG, Svelte 5 PWA, Docker.

FastAPI LLM RAG MCP GCP JS

IGVC | Autonomous Vehicle Competition

🔗 Blog Post

- Trained custom transformer BEV prediction models from Unreal Engine data.
- Design placed 2nd internationally 2023.

ROS Unreal Engine C++ PyTorch

SR AUV | Autonomous Underwater Vehicle

🔗 Blog Post

- Real-world tested end-to-end flying autonomously using PPO RL.
- Awarded best 2021 Capstone Project.

Unity C# TensorFlow TPU RL

SpaceX Hyperloop | Pod Competition

- Placed 8th at 2019 Pod Competition; built full-stack pod telemetry.

C++ Flask JS Protobuf CAN